PYTHON CODE:

```
# Store customer account IDs in a list
ids = [101, 205, 310, 415, 520]

# Take customer ID from the user
key = int(input("Enter Customer ID: "))

# --------- Linear Search ---------

# Check whether the ID exists in the list
if key in ids:
    print("Linear Search: ID Found")
else:
    print("Linear Search: ID Not Found")

# --------- Binary Search ---------

# Set the starting position
low = 0

# Set the ending position
high = len(ids) - 1

# Initially, ID is not found

found = False

# Continue searching while the range is valid
while low <= high:
    # Find the middle position
    mid = (low + high) // 2

    # Check if middle ID is the required ID

    if ids[mid] == key:
        found = True
        break

    # If key is greater, search the right half
    elif ids[mid] < key:
        low = mid +1
```

```
    # If key is smaller, search the left half
    else:
        high = mid - 1

# Display Binary Search result
if found:
    print("Binary Search: ID Found")
else:
    print("Binary Search: ID Not Found")
```

OUTPUT:

```
Enter Customer ID: 310

Linear Search: ID Found
Binary Search: ID Found
```